

Title: DATA VORTEX '26 — PHASE 2 ANALYTICAL SQL QUERIES

Event & Stage: Aaruush '26 | Round 1 (Phase 2)

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QUESTION 1

E3—Average Engagement by Platform

Challenge:

Calculate the average likes, shares, and comments for each platform.

Which platform generates the highest average total engagement?

SQL Query —

```
SELECT
    platform,
    COUNT(post_id) AS total_posts,
    ROUND(AVG(COALESCE(likes, 0)), 2) AS avg_likes,
    ROUND(AVG(COALESCE(shares, 0)), 2) AS avg_shares,
    ROUND(AVG(COALESCE(comments, 0)), 2) AS avg_comments,
    ROUND(AVG(COALESCE(likes, 0) + COALESCE(shares, 0) + COALESCE(comments, 0)), 2) AS avg_total_engagement
FROM posts
WHERE platform IS NOT NULL
    AND TRIM(platform) != ''
GROUP BY platform
ORDER BY avg_total_engagement DESC;
```

QUESTION 2

M1 — Which Locations Generate the Most Engagement?

Challenge:

Using both datasets, calculate the total engagement generated by users from each location. Return the location, number of posts, and total engagement. Rank locations from highest to lowest engagement

SQL Query —

```
WITH LocationEngagement AS (
```

```

SELECT
    COALESCE(u.location, 'Unknown') AS location,
    COUNT(p.post_id) AS post_count,
    SUM(COALESCE(p.likes, 0) + COALESCE(p.shares, 0) +
COALESCE(p.comments, 0)) AS total_engagement,
    ROUND(AVG(COALESCE(p.likes, 0) + COALESCE(p.shares, 0) +
COALESCE(p.comments, 0)), 2) AS avg_engagement_per_post
FROM users u
JOIN posts p ON u.user_id = p.user_id
GROUP BY COALESCE(u.location, 'Unknown')
)
SELECT
    location,
    post_count,
    total_engagement,
    avg_engagement_per_post,
    DENSE_RANK() OVER (ORDER BY total_engagement DESC) AS engagement_rank
FROM LocationEngagement
ORDER BY engagement_rank ASC;

```

QUESTION 3

H4 — Follower to Engagement Anomaly

Challenge:

Find users who have fewer than 5,000 followers but whose total post engagement places them among the top 10% of all users. These users may represent unusually high performing or suspicious accounts. This requires multiple levels of analysis.

SQL Query —

```

WITH UserAggregates AS (
    SELECT
        u.user_id,
        u.location,
        u.follower_count,
        COUNT(p.post_id) AS total_posts,

```

```

        SUM(COALESCE(p.likes, 0) + COALESCE(p.shares, 0) +
        COALESCE(p.comments, 0)) AS total_user_engagement
    FROM users u
    JOIN posts p ON u.user_id = p.user_id
    GROUP BY u.user_id, u.location, u.follower_count
),
RankedUsers AS (
    SELECT
        user_id,
        location,
        follower_count,
        total_posts,
        total_user_engagement,
        NTILE(10) OVER (ORDER BY total_user_engagement DESC) AS
engagement_decile
    FROM UserAggregates
)
SELECT
    user_id,
    location,
    follower_count,
    total_posts,
    total_user_engagement,
    engagement_decile
FROM RankedUsers
WHERE follower_count < 5000
    AND engagement_decile = 1
ORDER BY total_user_engagement DESC;

```